

ATOKA, OK, USA

WMO: 722051

Lat: 34.398N

Lon: 96.148W

Elev: 590

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	22.7	8.5	8.7	26.4	11.7	10.2	31.0	20.9	57.4	19.4	55.5	3.6	10	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	101.6	74.5	98.6	75.1	95.2	75.4	81.5	89.0	79.5	87.9	77.9	88.3	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	155.3	86.5	77.3	145.1	83.9	75.0	134.3	82.4	45.7	90.0	43.5	88.9	41.8	88.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.5	14.2	10.6	104.0	4.6	6.1	7.3	108.4	4.7	111.9	2.1	115.4	-1.2	119.8		
			9.7	81.0	4.1	2.1	6.7	82.5	4.3	83.8	2.0	85.0	-1.0	86.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	41.8	46.5	55.8	62.5	70.2	79.6	82.9	82.1	75.2	63.9	52.5	43.4
	DBStd	16.57	9.76	10.92	9.47	8.16	6.93	4.72	4.78	5.68	6.52	9.04	9.70	8.80
	HDD50	865	289	180	49	10	0	0	0	0	0	7	86	244
	HDD65	2915	718	524	308	138	32	0	0	0	5	133	386	671
	CDD50	5654	36	82	230	384	625	889	1019	996	755	439	161	38
	CDD65	2232	0	5	24	63	193	439	554	531	310	100	12	1
	CDH74	24604	7	31	181	570	1740	4776	6787	6238	3186	1007	76	5
	CDH80	12139	0	4	24	115	581	2319	3805	3439	1490	357	5	0
Wind	WSAvg	4.8	5.3	6.2	6.5	6.2	5.1	4.6	3.5	3.2	3.4	4.2	5.2	4.8
Precipitation	PrecAvg	47.1	2.4	2.3	4.1	5.0	5.8	4.8	3.1	2.9	4.9	4.6	2.8	3.1
	PrecMax	66.3	7.9	5.4	8.3	10.7	10.9	9.3	9.1	7.4	9.5	12.7	8.5	8.2
	PrecMin	29.5	0.0	0.0	0.8	0.7	0.6	1.6	0.0	1.0	1.3	0.9	0.5	0.1
	PrecStd	9.3	1.9	1.6	2.0	2.2	3.2	2.4	2.5	1.5	2.4	3.0	2.2	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	77.5	82.2	88.4	92.5	99.5	106.1	108.5	99.4	90.6	80.8	73.0
		MCWB	56.4	61.8	65.7	69.8	74.4	75.6	74.6	73.6	71.9	70.8	64.3	60.7
	2%	DB	69.7	72.9	79.1	82.3	90.1	96.8	102.0	102.2	95.0	88.1	76.5	68.2
		MCWB	55.5	60.3	63.7	66.6	73.6	75.5	74.9	74.3	73.5	70.8	61.5	58.0
	5%	DB	64.0	69.7	74.8	80.6	86.3	93.2	98.7	98.7	91.1	84.0	72.6	63.3
		MCWB	54.6	59.1	62.1	65.6	72.1	74.1	75.5	75.2	72.3	68.1	61.3	54.2
	10%	DB	59.4	64.5	72.1	77.2	82.2	90.8	95.2	94.7	88.5	80.6	69.5	57.5
		MCWB	48.9	56.5	61.0	64.8	70.1	75.6	76.2	75.4	71.7	66.2	60.8	49.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	66.2	69.1	73.0	78.3	82.2	83.8	83.9	79.4	75.6	68.5	65.5
		MCDB	67.2	73.6	77.5	82.0	89.3	89.8	94.3	93.3	87.6	87.7	73.8	70.9
	2%	WB	60.3	63.9	66.5	70.9	75.7	80.6	81.4	81.5	76.7	72.8	66.3	61.1
		MCDB	65.5	69.3	75.3	79.3	85.3	87.8	89.5	89.2	86.5	83.1	72.7	65.0
	5%	WB	55.3	61.7	64.9	69.0	74.0	78.5	78.9	79.0	74.8	71.2	64.2	56.4
		MCDB	62.4	66.8	72.2	76.6	82.9	87.7	88.6	87.9	85.7	79.9	69.2	60.6
	10%	WB	50.5	57.7	63.1	67.2	72.3	77.2	77.3	77.5	73.5	69.1	61.4	52.2
		MCDB	58.4	64.7	69.0	74.6	80.5	87.6	89.1	88.6	84.3	75.3	67.3	57.0
Mean Daily Temperature Range	5% DB	MDBR	22.3	21.8	21.9	22.4	20.0	20.0	21.7	21.1	22.8	23.1	23.1	19.9
		MCDBR	30.8	27.0	25.4	24.6	23.0	23.0	26.2	26.0	25.5	26.3	24.7	27.3
		MCWBR	19.2	16.7	13.7	12.5	10.2	7.8	6.8	6.7	8.4	11.6	14.6	17.8
	5% WB	MCDBR	25.0	22.8	20.1	19.4	19.0	19.6	21.6	21.3	21.4	21.8	20.0	21.5
MCWBR		18.9	16.4	12.6	12.5	10.1	9.1	9.6	9.5	8.4	11.4	15.7	17.6	
Clear-Sky Solar Irradiance	taub	0.309	0.318	0.355	0.386	0.424	0.433	0.442	0.446	0.398	0.352	0.331	0.309	
	taud	2.491	2.484	2.405	2.328	2.286	2.298	2.313	2.283	2.371	2.487	2.486	2.499	
	Ebn at Noon	286	296	291	285	274	271	268	265	273	279	275	279	
	Edh at Noon	27	30	36	40	42	42	41	42	36	30	27	25	
All-Sky Solar Radiation	RadAvg	852	1011	1351	1651	1794	2045	2088	1912	1592	1225	907	736	
	RadStd	88	150	95	124	181	225	123	121	155	178	113	94	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135

Nomenclature: See separate page